PLATFORM DOMAIN · DISCOVERY AND VALIDATION

Map the nine anatomy elements onto this domain's agent

Version 3 · 2026-10-19 Prepared by Q. Nkwain Ninying Autonomy level today: R0

This page maps the agent's anatomy using the standard components of an agentic system, taken from the revised Reliability Agent Proposal. The last column shows what implements each component. It stays a draft until the manager approves it below.

What changed

Version 3: replaced "Runs on" with "Implemented by": model, data store, runtime code, or prompt and config, with a note where two of them share the work. Removed the anatomy-term column. Memory now names the context window as working memory.

Version 2:

- Replaced the nine body-part terms with ten standard engineering components, so people and Claude read the design the same way.
- Perception is system input (logs, schedules, errors, incidents), not a user talking to the agent.
- Memory starts with an audit of the existing logging, and the agent's own memory is kept separate from the system's logs.
- No separate voice: reports and answers are produced through the tool layer.
- Every component is built together, small, with evaluation alongside, rather than some parts later.

The loop

`perceive` → `reason` → `plan` → `act` → `verify` → `remember`

Guardrails sit around every step and are enforced by the runtime, not left to the model.

The components

COMPONENT	ROLE IN THIS AGENT	IMPLEMENTED BY
Reasoning engine (LLM core)	Interprets logs and signals, forms hypotheses, and decides what to do next.	MODEL
Planner	Builds each investigation workflow at runtime from the evidence it finds, and re-plans when a step fails or new evidence appears.	MODEL plan run by the runtime
Memory	Working memory for the current incident (the context window). Long-term memory for confirmed past incidents, runbooks and baselines. Kept separate from the system's own logs.	DATA STORE model summarizes and matches
Perception layer	System inputs read in place: activity logs, schedules, errors, health checks, performance and capacity data, inventory and lineage, tickets, deploy history. Read-only; nothing is copied.	RUNTIME CODE
Tool layer	Typed tools the planner may call: read-only queries, saving a report, drafting a ticket. No tool fixes, restarts or changes anything.	RUNTIME CODE
Orchestration runtime	Runs the plans and supervises the loop: triggers, grouping related alerts into one incident, state, retries, timeouts, follow-ups, routing reports to owners.	RUNTIME CODE
Agent specification	The system prompt and boundaries: reliability-only scope, definition of done for a report, severity and priority rules, the report schema.	PROMPT + CONFIG
Guardrails and governance	Least-privilege read-only credentials, a tool allowlist, evidence for every claim, step and cost budgets, audit logging, no automatic edits to its own memory.	RUNTIME CODE enforced outside the model
Response generation	Writes the report and answers "why did you flag this?", with stated confidence. Says "I don't	MODEL

COMPONENT	ROLE IN THIS AGENT	IMPLEMENTED BY
	know" when evidence is thin. Delivered through the tool layer.	
Verification and feedback loop (evals)	After a person fixes something, re-runs the check and records whether the problem went away. Feeds evaluation and, once a person confirms it, long-term memory.	RUNTIME CODE model reads unclear results

What comes first

- Audit the existing logging: does it capture enough depth and detail for the agent to diagnose any issue? This defines what perception and memory can rely on.
- Build every component together, each in its smallest useful form, with evaluation from the first run.
- At R0 every component above is a definition. Nothing is built or running yet.

Manager review

● APPROVED · VERSION 3

Approved by: K.D.
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